How We Calibrated WOFOST to Texcoco Farm Yields

What was done, why, and how to read the results

Gradient Descent Farmers — companion to the technical report

October 18, 2026

1. In one paragraph

Before calibration, the crop simulator predicted yields 2.5 to 5 times higher than what Texcoco farmers report to the Mexican government (6.3 vs 2.6 t/ha for maize, 6.1 vs 1.8 for barley, 9.8 vs 1.9 for faba bean). We picked a few model settings ("knobs") with a clear physical meaning and tried many combinations. For each crop we kept the combination whose simulated yields came closest to the observed yields of 2006–2021. We then checked it on 2022–2025, years the search never saw. After calibration the simulated averages are within about 0.2 t/ha of the observed ones, and the error drops from 4–8 t/ha to under 1 t/ha. The model still cannot tell good years from bad ones reliably, so calibrated yields should be read as realistic averages and ranges, not forecasts.

2. What "calibration" means here

WOFOST, the crop model we use, does not learn from yields. It computes a yield day by day from weather, soil and about a hundred crop parameters (see the companion *What Kind of Model Is WOFOST?*). Those parameters come from field experiments in Europe: they describe modern varieties grown with good weed, pest and fertilizer management. Texcoco's rainfed farms are different, so the model's yields are attainable yields, not farm yields.

Calibration means adjusting a small number of those parameters so that the model's output matches observed data for the place we care about. It is a curve fit, but of a few physically meaningful numbers inside the model, not of the yield itself.

Why not just multiply the yields by a correction factor? The farmer notebook did exactly that: it multiplied every maize yield by 0.40, barley by 0.28 and faba bean by 0.18. That fixes the average but nothing else:

- the model still believes fertilizer, frost and drought act on a 6–10 t/ha crop, so the size of every scenario effect ("20% less rain", "no fertilizer") is wrong;
- the same factor is applied in every year, whatever happened in that year.

Changing parameters *inside* the model lets the lower yield interact with water, nitrogen and frost the way a real low-input crop would.

3. The ingredients

3.1 Observed yields: SIAP

SIAP (Servicio de Información Agroalimentaria y Pesquera) publishes, every year and for every municipality, the area sown, the area harvested, the area lost and the production of each crop. We

keep the rows for Texcoco (state 15, municipality 99), the spring–summer cycle and rainfed land (`farm_tool.siap_texcoco`). From them we use the **yield per sown hectare**:

$$y^{\text{obs}} = \frac{\text{production (t)}}{\text{sown area (ha)}}. \tag{1}$$

Dividing by the *sown* area (not the harvested area) counts fields that were lost to frost or drought as zero yield. That is what a failed season looks like in the model, so the two are comparable. In 2011 and 2019, for example, 51% and 64% of the maize area was lost, and the yield per sown hectare falls to 1.0 and 0.8 t/ha.

Crop	Years with data	Mean 2006–2021	Mean 2022–2025	Sown area, 2022–2025
Maize	2005–2025	2.56 (16 years)	2.98	900–1,400 ha
Barley	2008–2025	1.78 (14 years)	1.55	4–9 ha
Faba bean	2005–2019, 2022–2025	1.94 (14 years)	1.17	10–12 ha

Table 1: The observations used (yields in t/ha per sown hectare). Barley has no data before 2008 and faba bean none for 2020–2021.

3.2 Simulated yields

The simulator (`wofost_rotations.py`) runs four rotations (maize every year; maize–barley; maize–faba bean; maize–barley–faba bean) from 2005 to 2025. Every *phase* of a rotation is simulated: maize–barley is run once starting with maize and once starting with barley, so every crop appears every year. Seasons are chained, so the water and nitrogen left by one crop start the next one.

The yield we compare is the **water- and nitrogen-limited yield** (rainfed, with fertilizer, with the grain-N constraint), the most realistic of the three production levels the simulator computes. Because SIAP reports one number per crop and year for the whole municipality, we average the simulated yield of each crop over all rotations and phases in which it is grown that year:

$$y^{\text{sim}}_{c,t} = \frac{1}{n_{c,t}} \sum_{\text{rotation, phase growing } c \text{ in year } t} Y_{\text{rotation, phase},t}. \tag{2}$$

This is the same comparison as `farm_tool.farm_gap`. The first year, 2005, is a "spin-up": it starts from assumed soil water and nitrogen, so it is left out.

3.3 Fit years and holdout years

A model with enough knobs can match any short series, including its noise, and then fail on new years. This is called *overfitting*. To check for it, the years are split:

- **Calibration (fit) years, 2006–2021**: the only years the search looks at when choosing knob values.
- **Holdout years, 2022–2025**: never used to choose anything. They are reported at the end as an honest test. Four years is little, so the holdout tells us about the level more than about year-to-year agreement.

4. How a fit is scored

For one crop and the n years with an observation, let $e_t = y_t^{\text{sim}} - y_t^{\text{obs}}$ be the error in year t. We report four numbers:

$$\text{Level:} \quad \bar{y}^{\text{sim}} \text{ vs } \bar{y}^{\text{obs}} \quad \text{(the two means; their ratio is the old correction factor)} \tag{3}$$

$$\text{Bias:} \quad \bar{e} = \frac{1}{n}\sum_t e_t \quad \text{(positive = model too high)} \tag{4}$$

$$\text{RMSE:} \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_t e_t^2} \quad \text{(typical size of the error, in t/ha)} \tag{5}$$

$$\text{Correlation:} \quad r = \frac{\sum_t (y_t^{\text{sim}} - \bar{y}^{\text{sim}})(y_t^{\text{obs}} - \bar{y}^{\text{obs}})}{\sqrt{\sum_t (y_t^{\text{sim}} - \bar{y}^{\text{sim}})^2}\sqrt{\sum_t (y_t^{\text{obs}} - \bar{y}^{\text{obs}})^2}} \tag{6}$$

r runs from -1 to 1. It measures whether good and bad years line up (1 = perfectly, 0 = no relation), whatever the level.

The search minimizes RMSE. It is useful to know that

$$\text{RMSE}^2 = \bar{e}^{\,2} + \text{SD}(e)^2, \tag{7}$$

so RMSE punishes both a wrong level ($\bar{e}$) and wrong year-to-year swings ($\text{SD}(e)$). There is a trap here: when the model's swings do not line up with the observed ones, the cheapest way to lower RMSE is to make the model swing *less*. Section 7 shows how this trap showed up and how it was avoided.

5. Diagnosis: why is the default model so far off?

Before fitting anything, we changed one assumption at a time and reran all rotations (`python -m src.crop_models.calibration --sensitivity`). Table 2 summarizes the results.

Change from the default	Maize	Barley	Faba bean	Maize r	Maize ended by frost
None (default)	6.34	6.15	9.79	0.36	69%
Phenology ×0.8 (shorter cycle)	9.53	5.56	9.80	0.40	25%
Phenology ×0.65	10.72	4.24	8.22	0.39	6%
Photosynthesis ×0.5	3.44	2.77	2.53	−0.01	69%
Fertilizer N recovery 0.25 (not 0.5)	4.12	4.97	9.77	0.41	69%
Soil N mineralization halved	4.53	3.83	9.74	0.40	69%
Frost kills at −2 °C (not 0 °C)	6.50	6.15	9.89	0.34	44%
Soil 50 cm deep (not 100 cm)	6.37	6.16	9.79	0.35	69%
SIAP (observed)	2.56	1.78	1.94		

Table 2: One-at-a-time sensitivity: mean simulated yield 2006–2021 (t/ha) over the years with SIAP data.

What the table and the model's internal variables tell us:

1. **The crop parameters describe high-yield European crops.** Without water or nitrogen limits ("potential" yield) the default simulates 11.2 t/ha of maize, 12.0 of barley and 10.0 of faba bean. Rain barely matters on average: the 100 cm soil stores 236 mm of plant-available water and May–October brings about 770 mm on average, so the rainfed yields are the same as the potential ones.

2. **Nitrogen is the only brake, and only on maize and barley.** Default maize takes up about 150 kg of N/ha: 80 kg from fertilizer (160 kg/ha applied, of which the model lets the crop recover half) and about 64 kg per season from the soil's own organic matter (mineralization). Halving either source lowers maize by about 2 t/ha. Faba bean is a legume that takes 80% of its nitrogen from the air, so nothing reduces it.

3. **Phenology and frost.** NASA POWER weather comes from a grid cell whose average elevation is 2,633 m, about 380 m above the fields at El Batán. Air cools about 6.5 °C per km of altitude, so the weather the model sees is roughly 2.5 °C colder than the fields. Maize development is driven by accumulated temperature, so it is slow. The default maize flowers about 87 days after sowing (plausible for Valles Altos hybrids) but then needs about 100 more days to fill its grain. In 69% of seasons the first frost (a night at or below 0 °C) ends the crop before it matures, and when it matures the median date is December 5. This is why frost dates drive the model's good and bad years.

4. **Frost threshold and soil depth matter little.** Neither changes the mean by more than 0.2 t/ha.

5. **Photosynthesis is a strong level lever but erases the year pattern.** Halving it brings all three crops much closer to SIAP, but maize's correlation drops from 0.36 to about 0.

Why does continuous maize almost fail in 2023? SIAP reports 2023 as Texcoco's best maize year (4.0 t/ha). Averaged over rotations, the default model also has a good 2023. The near-failure (0.01 t/ha) only happens when maize follows maize:

1. the 2022 maize was sown late (June 15) and kept transpiring until January 5, 2023;
2. on March 1, 2023 the soil therefore held under 10 mm of water the plant could use, against 130–200 mm when barley or faba bean had been grown in 2022 (no rain falls in the dry season to refill it);
3. the 2023 crop was sown on May 15, after about 24 mm of early-May rain, and then June was almost rainless (a 28-day dry spell; 2023 had the driest May–October of 2005–2025 in the POWER data, 478 mm);
4. with no water to take up, the seedlings could not grow; their leaf area shrank from 0.05 to 0.003 and never recovered. WOFOST has no "re-sowing", which a farmer would do.

A shorter maize cycle avoids this: with TSUM ×0.8 the same season yields 9.9 t/ha.

6. The knobs

We chose four knobs, each tied to a specific mechanism found in Section 5. The defaults reproduce the uncalibrated model exactly.

Knob	WOFOST parameters changed	Default	What it represents
Phenology scale s	TSUM1, TSUM2 $\times s$	1	A shorter or longer-cycle variety
Photosynthesis scale p	AMAX_REF, AMAX_SLP, EFFTB $\times p$	1	Losses WOFOST does not simulate: weeds, pests, diseases, uneven plant stands, landrace genetics
Fertilizer N recovery f	share of each N dose the crop can access	0.5	Lower doses than the 160 kg/ha plan, losses to leaching and volatilization
Soil mineralization m	N released per season = $m\times$ soil total N	0.5%	Fertility of the soil organic matter (one value for the site)

Table 3: The calibration knobs. The first three are set per crop; mineralization is shared by all crops.

Phenology (TSUM). WOFOST tracks a development stage D that goes from 0 at emergence to 1 at flowering and 2 at maturity. Each day it grows by the day's effective temperature divided by a thermal-time requirement:

$$D_{t+1} = D_t + \frac{T_{\text{eff}}(t)}{\text{TSUM1}} \ (D < 1), \qquad D_{t+1} = D_t + \frac{T_{\text{eff}}(t)}{\text{TSUM2}} \ (D \geq 1), \tag{8}$$

where, for maize, $T_{\text{eff}} = \min(\max(T - 6, 0), 24)$ in °C day. Multiplying TSUM1 and TSUM2 by $s < 1$ makes every stage arrive sooner. Table 4 shows what this means in days after sowing at Texcoco.

Crop	Phenology scale	Days to flowering	Days to end of season	Seasons reaching maturity
Maize	1.0 (default)	87	191	31%
	0.8	72	155	75%
	0.6	56	116	94%
Barley	1.0 (default)	64	121	100%
	0.8	53	97	100%
Faba bean	1.0 (default)	58	146	75%
	0.8	48	118	100%

Table 4: Simulated phenology, medians over 2006–2021 (maize median sowing date: May 27). A season ends at maturity, at the first frost, or on February 28.

Photosynthesis. In WOFOST 8.1 the maximum CO_2 uptake of a leaf depends on its nitrogen content: $A_{\max}$ = CO2 factor × temperature factor × min(AMAX_REF, AMAX_SLP · SLN), and EFF is the light-use efficiency of a leaf in dim light. Scaling all three by p scales each leaf's CO_2 uptake, and so the crop's daily sugar production, by p. Everything downstream (leaf growth, roots, grain, N demand) responds to it, which is why this works better than multiplying the final yield.

Nitrogen. Each fertilizer dose of F kg N/ha makes $f \cdot F$ available to the crop; the rest is lost. The soil also releases $m \times 12{,}709$ kg N/ha (IGBP total N in 0–100 cm) per season, spread evenly over the first 150 days from March 1. With the defaults that is $0.005 \times 12{,}709 \approx 64$ kg; with $m = 0.25\%$, about 32 kg.

How the knobs reach the model. The calibration is passed to `simulate_rotations(..., calibration=...)` as a dictionary. Crop parameters are applied as PCSE *overrides* on the

parameter provider of each season. (Editing the crop parameter dictionary directly does not work: PCSE reloads the variety's parameters when the crop is sown, which silently undid the first attempt.) When no calibration is passed, the code takes exactly the old path. We checked this by rerunning the default baseline and the `no_fertilizer` and `canicula50` scenarios: the outputs are identical to the saved CSVs.

7. The search

7.1 Grid search, one crop at a time

A **grid search** lists a few values for each knob and simulates every combination (a "candidate"). For each candidate we simulate the rotations that contain the crop, compute the crop's RMSE over 2006–2021, and keep the candidate with the lowest RMSE. Figure 1 shows the loop.

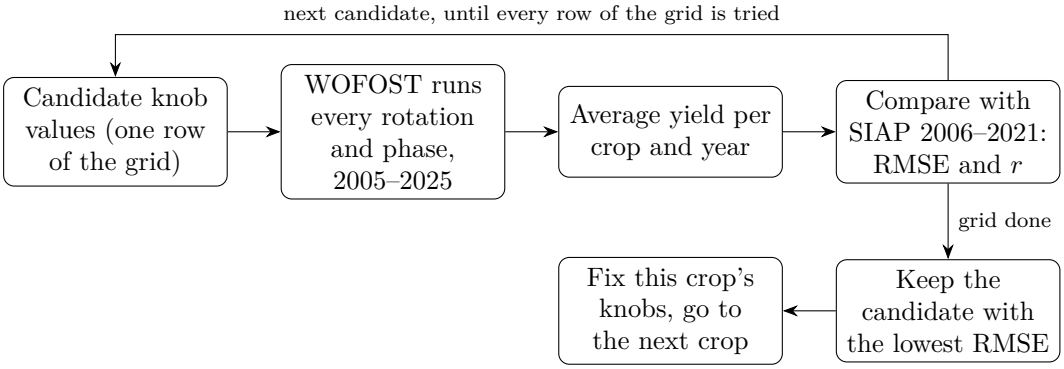


Figure 1: The calibration loop. The holdout years 2022–2025 never enter it.

Trying every combination of all knobs for all three crops at once would take days, so the crops are fitted one at a time ("coordinate search"):

1. **Barley** (24 candidates): phenology $\{0.8, 0.9, 1\}$, photosynthesis $\{0.4, 0.5, 0.6, 0.8\}$, N recovery $\{0.25, 0.5\}$;
2. **Faba bean** (15): phenology $\{0.8, 0.9, 1\}$, photosynthesis $\{0.4, 0.45, 0.5, 0.55, 0.6\}$; N recovery stays at 0.5 because the crop fixes its own N;
3. **Maize** (54), with barley and faba bean at their fitted values: phenology $\{0.8, 0.9, 1\}$, N recovery $\{0.1, 0.2, 0.25, 0.3, 0.4, 0.5\}$, mineralization $\{0.25\%, 0.375\%, 0.5\%\}$; photosynthesis stays at 1 (see 7.2).

Maize goes last because it depends on its partners (it inherits the water and nitrogen they leave) more than they depend on it: at the fitted photosynthesis barley is hardly N-limited, and faba bean fixes its N. To save time, the search only simulates the water- and N-limited level, one of the simulator's three production levels. The 93 candidates run on 8 processor cores in about 9 minutes.

Ties. Several candidates can score the same (for example, barley's N recovery has no effect once its photosynthesis is low enough that nitrogen no longer limits it). RMSE and r are compared to three decimals; equal RMSE goes to the higher correlation, and then to the candidate closest to the defaults, measured as $\sum_k |v_k/\text{default}_k - 1|$. So barley keeps the default N recovery of 0.5.

7.2 Limits on the grid, and what earlier attempts taught us

The final grid is the fourth version. Each earlier attempt revealed something:

Attempt	What the search chose for maize	What we learned and changed
1	Phenology ×0.6, photosynthesis ×0.6, N recovery 0.25: all at the edge of the grid	×0.6 means flowering 56 days after sowing, a variety not grown in Texcoco. Most likely the search used it to make yields swing less (the RMSE trap of Section 4). Phenology was limited to ×0.8–1.
2	Default phenology and photosynthesis, N recovery 0.15 (lowest value tried): mean 3.31, RMSE 1.09, r 0.41	Nitrogen supply, not photosynthesis, is the right lever for maize: lowering photosynthesis matched the level but dropped r to about 0.1, and below ×0.5 yields drop abruptly. The grid was extended to lower N recovery.
3	N recovery 0.05 (again the lowest value), even after refitting maize with calibrated partners: mean 3.19, RMSE 1.00	The calibrated barley and faba bean are smaller and use less N, so more N is left for the next maize. Fertilizer alone cannot bring maize down: the soil supplies too much. Mineralization was added as a knob.
4 (final)	Default phenology and photosynthesis, N recovery 0.2, mineralization 0.25%: mean 2.35, RMSE 0.71, r 0.39	An interior optimum (not on the edge for any knob that matters).

Table 5: The four attempts. A choice on the edge of the grid means the best value may lie outside it.

Two limits are deliberate and come from agronomy, not from the fit:

- **Phenology at least** ×0.8 for all crops, to keep crop calendars realistic.
- **Photosynthesis at least** ×0.4 for barley and faba bean. Below that the crops fail to establish: barley yields fall from 1.6 t/ha at ×0.5 to 0.7–1.2 t/ha at ×0.4 (Figure 5).

8. Results

Crop	Phenology	Photosynthesis	Fertilizer N recovery
Maize	×1.0	×1.0	0.20
Barley	×0.8	×0.5	0.5
Faba bean	×1.0	×0.45	0.5
Site	Soil mineralization 0.25% of total N per season (about 32 kg N/ha, default 64)		

Table 6: The fitted knobs, stored as `CALIBRATED` in `src/crop_models/calibration.py`.

		Fit years 2006–2021			Holdout 2022–2025		
Crop	Model	sim / obs (t/ha)	RMSE	r	sim / obs (t/ha)	RMSE	r
Maize	default	6.34 / 2.56	4.24	0.36	7.63 / 2.98	4.66	0.92
	calibrated	2.35 / 2.56	0.71	0.39	2.74 / 2.98	0.89	−0.25
Barley	default	6.15 / 1.78	4.37	−0.08	5.81 / 1.55	4.28	−0.21
	calibrated	1.65 / 1.78	0.28	−0.11	1.30 / 1.55	0.41	0.21
Faba bean	default	9.79 / 1.94	7.87	−0.11	9.59 / 1.17	8.45	−0.55
	calibrated	1.73 / 1.94	0.61	−0.18	1.73 / 1.17	0.60	−0.27

Table 7: Before and after. Holdout correlations rest on four years and should not be over-read.

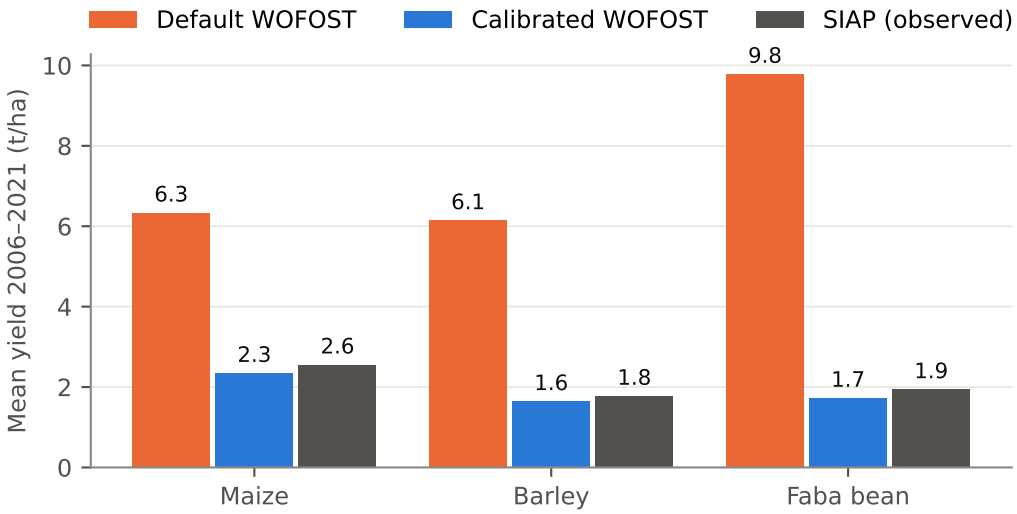


Figure 2: Mean yield over the fit years: the calibrated model (blue) now sits next to SIAP (gray).

How to read the results.

- **The level is fixed.** In the fit years the calibrated means are 0.1–0.2 t/ha below SIAP for all three crops, and the RMSE falls 6 to 15 times. In the holdout years, which the search never saw, the RMSE falls 5 to 14 times. The old `farm_gap` correction factors (0.18–0.40) become 1.02–1.10, so the notebook no longer needs them.
- **Year-to-year agreement is not improved.** Maize keeps $r \approx 0.4$ because the model reproduces the 2011 loss (late sowing and an October 23 frost). It also invents a poor 2010 (frost on November 4) and misses SIAP's 2019 loss (Figure 3). Barley and faba bean show no agreement before or after.
- **Faba bean's holdout bias (+0.55 t/ha)** comes from the data: SIAP faba bean yields fall steadily from 2.6 t/ha in 2005–2009 to 1.2 in 2016–2025, a trend no weather-driven model can reproduce.
- **2023.** Averaged over rotations, calibrated maize gives 2.8 t/ha in 2023 against SIAP's 4.0. Continuous maize still fails that year, for the reasons of Section 5.

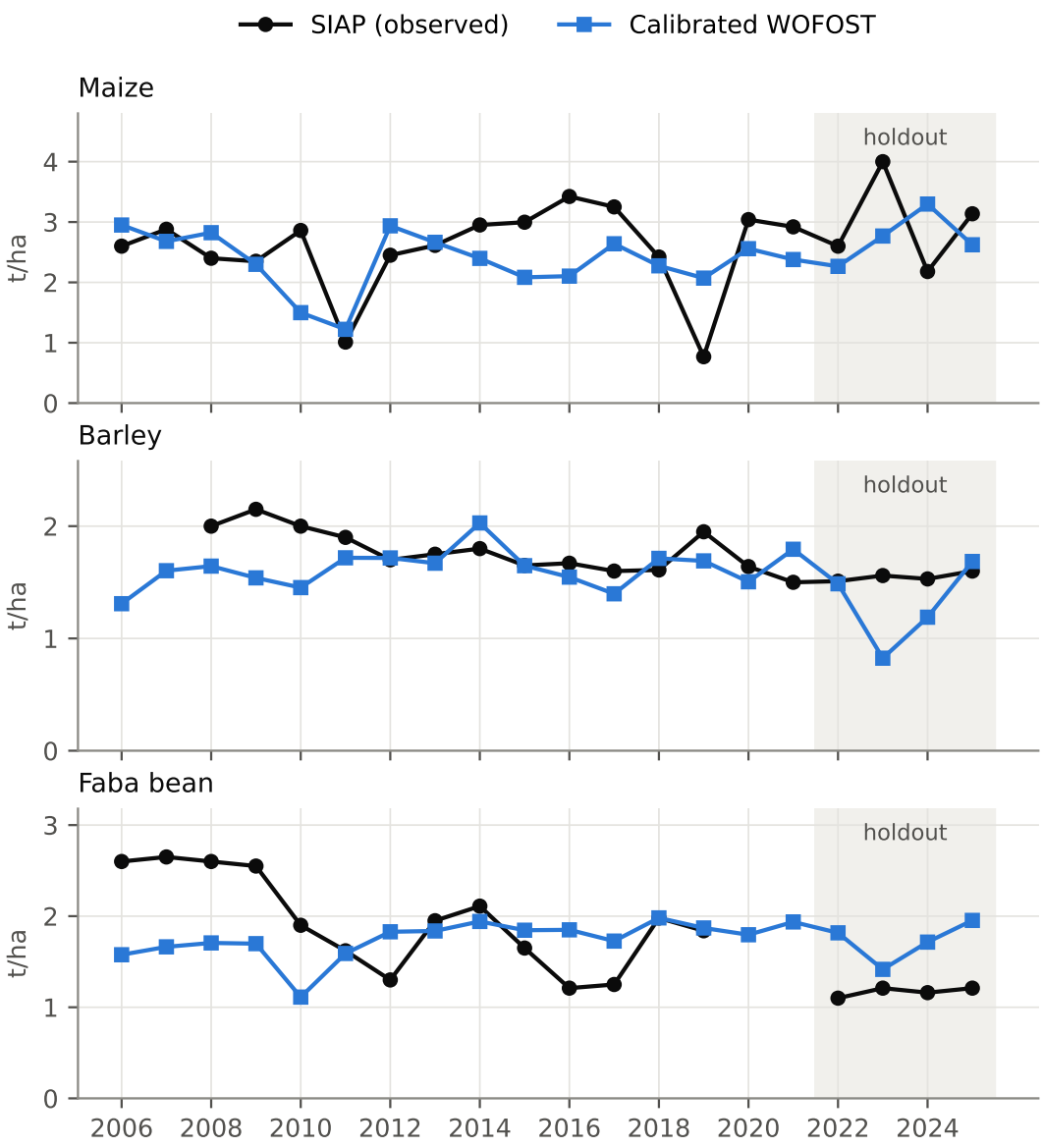


Figure 3: Year by year: calibrated WOFOST (blue squares) against SIAP (black dots). The shaded years are the holdout. Missing black dots are years without SIAP data.

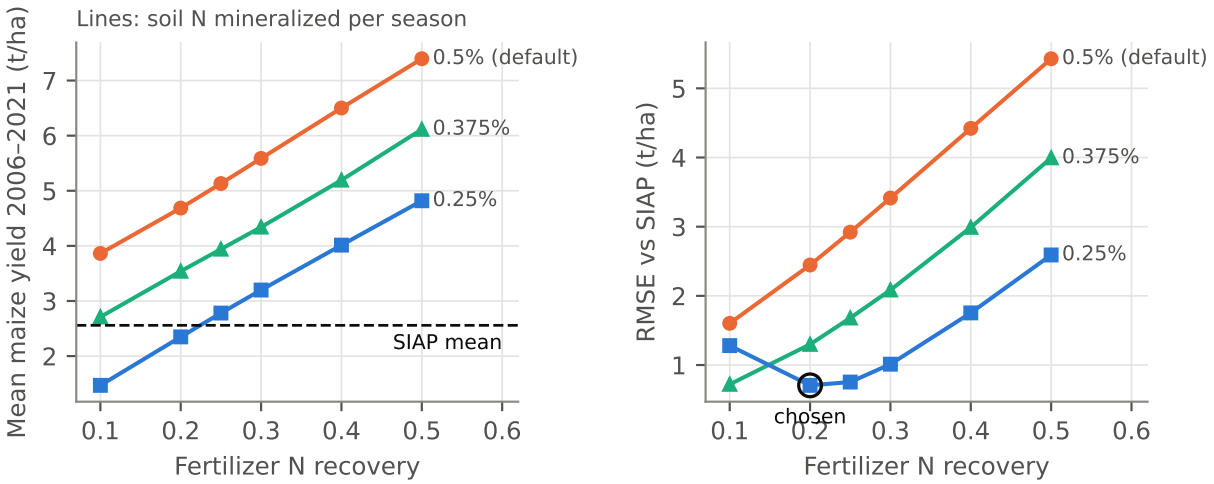


Figure 4: The maize step of the search (phenology ×1). Left: mean maize yield for each combination of fertilizer N recovery and soil mineralization; the dashed line is the SIAP mean. Right: the RMSE of the same candidates; the circled point was chosen. Several combinations reach the observed level.

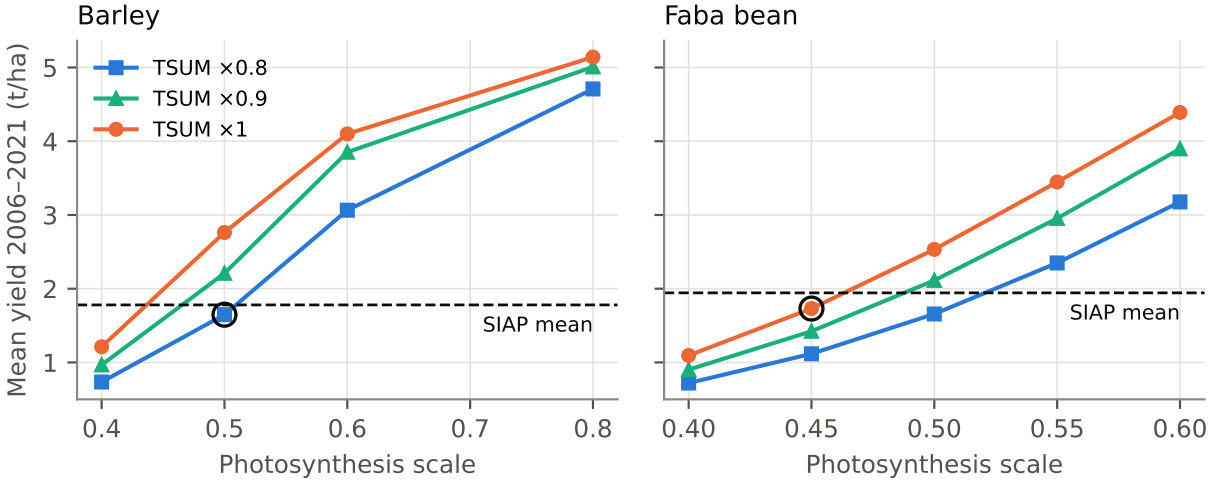


Figure 5: The barley and faba bean steps (fertilizer N recovery 0.5). Each line is a phenology scale; the circled point was chosen; the dashed line is the SIAP mean. Yields climb steeply with photosynthesis.

9. Limits you should know about

1. Fertilizer and soil nitrogen cannot be told apart. SIAP only tells us how much a field yields, not where its nitrogen came from. Figure 4 shows this: several pairs of (fertilizer recovery, mineralization) produce the observed maize level. The two best candidates are almost tied:

Mineralization	Fertilizer N recovery	Mean maize (t/ha)	RMSE	r
0.25%	0.20	2.35	0.706	0.39
0.375%	0.10	2.71	0.721	0.40

This matters for scenarios, because it decides how much fertilizer matters:

Mean yield, all years 2006–2021 (t/ha)	Default model		Calibrated model	
	with fertilizer	without	with fertilizer	without
Maize	6.34	2.03	2.35	0.29
Barley	6.15	3.90	1.62	1.45
Faba bean	9.81	9.77	1.75	1.75

In the calibrated model, unfertilized maize collapses to 0.3 t/ha, which is probably too harsh. Results of the `no_fertilizer` and `half_fertilizer` scenarios should be treated with care until on-farm fertilizer trials are available to split the nitrogen supply.

2. Barley and faba bean are close to an establishment cliff. Their fitted photosynthesis sits just above the point where they fail to establish (Figure 5). Very dry scenarios may therefore exaggerate their failures; for example, calibrated barley drops to 0.8 t/ha in the dry 2023.

3. Phenology is not pinned by observations. We have no observed flowering or maturity dates, so phenology was bounded by agronomic judgment rather than fitted to dates. The fit kept the default maize cycle, which is still cut short by frost in 69% of seasons.

4. The observations are coarse. SIAP numbers are municipal estimates, often rounded (2.6, 2.4), and recent barley and faba bean figures describe only 4–12 hectares.

5. One point represents the whole municipality. One POWER grid cell and one soil point; real fields differ in soil depth (e.g. *tepetate* layers), sowing date and management.

10. How to use it and reproduce it

Everything is opt-in; nothing changes for existing commands. From the repository root:

```
# One-at-a-time diagnostics, Table 2 (about 2 min)
python -m src.crop_models.calibration --sensitivity
# The search (about 9 min on 8 cores)
python -m src.crop_models.calibration --fit
# Calibrated baseline -> data/rotations_calibrated/ (other scenarios with --scenarios)
python -m src.crop_models.calibration --simulate
python -m src.crop_models.calibration --simulate --scenarios baseline rain80
# The figures of this document
python docs/technical_report/make_calibration_figures.py
# Unit tests, including tests/test_calibration.py
python -m unittest discover -s tests
```

Files produced in `data/rotations_calibrated/`:

- `baseline.csv` (and any other scenario requested): the same columns as the default runs, with the scenario named `baseline_calibrated`, so default and calibrated runs can be loaded together;
- `calibration.json`: the knobs used;
- `fit_history.csv`: every candidate tried in the search and its scores.

In the farmer notebook, this compares the calibrated run with SIAP; its correction factors are close to 1:

```
calibrated = load_seasons(["data/rotations_calibrated/baseline.csv"])
farm_gap(calibrated, siap, scenario="baseline_calibrated")
```

If the knobs are refitted, copy the printed values into `CALIBRATED`.

11. Glossary

Calibration Adjusting a few model parameters so the model's output matches local observations.

Candidate One combination of knob values tried by the search.

Coordinate search Fitting one group of knobs (here, one crop) at a time while keeping the others fixed.

Correlation r Whether simulated and observed good and bad years line up; ignores the level.

Grid search Trying every combination of a short list of values for each knob.

Holdout Years kept out of the search and used only to test the result.

Mineralization Release of plant-available nitrogen from soil organic matter by microbes.

N recovery The share of applied fertilizer nitrogen that the crop can actually take up.

Overfitting Matching the fit years' noise so closely that new years are predicted badly.

Phase Where in its sequence a rotation starts; every phase is simulated so every crop appears each year.

Phenology The timing of crop development (emergence, flowering, maturity).

RMSE Root mean square error: the typical size of the difference between simulation and observation, in t/ha.

SIAP Mexico's agri-food and fisheries information service, source of the observed yields.

Spin-up The first simulated year, discarded because it starts from assumed soil conditions.

TSUM Thermal time (degree-days) a crop needs to reach flowering (TSUM1) or maturity (TSUM2).

Yield per sown hectare Production divided by the area sown, so lost fields count as zero yield.